Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam C · Form 2: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. A hanging m1 = 5 kg is connected by an ideal rope to m2 = 2 kg on a 30° incline. An extra vertically downward force of 0 N acts on m2. m2 slides uphill with $\mu_k = 0.1$. Find acceleration with uphill positive for m2. Use g = 9.8 m/s².

(A) 10.715 m/s²

(B) 2.679 m/s²

(C) 5.358 m/s²

(D) -5.358 m/s²

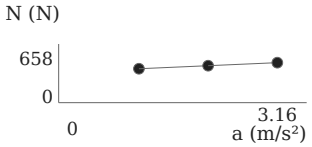
2. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.16, 558); (2.16, 608); (3.16, 658). Which horizontal setting produces a reading of 583?

(A) 1.66

(B) 14.16

(C) 26.66

(D) -10.84



3. Object A, whose mass is 8 times B's, collides with B. Ignore other horizontal forces during contact. Which comparison of force magnitudes and acceleration magnitudes is correct?

(A) Forces are equal; A's acceleration is 8 times B's.

(B) Different masses make both comparisons impossible.

(C) Forces are equal in magnitude; B's acceleration magnitude is 8 times A's.

(D) A exerts 8 times the force; accelerations are equal.

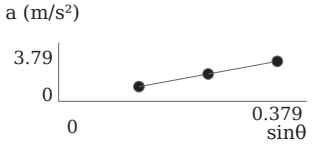
4. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1392, 1.392); (0.2592, 2.592); (0.3792, 3.792). Set the horizontal origin at the first measurement. What is the new vertical intercept?

(A) 1.392

(B) 1.949

(C) 2.506

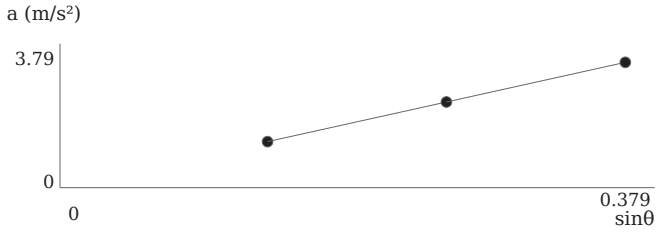
(D) 0.8352



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Release a block on a frictionless ramp and record downhill acceleration. Use g=10 m/s². sinθ -> a (m/s²): (0.1392, 1.392); (0.2592, 2.592); (0.3792, 3.792).

a. For the ramp tilt experiment setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Exchange the axes. Relate the new slope and its units to the original slope. **(3 pt)**

6. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.16, 21.16); (2.16, 22.16); (3.16, 23.16).

a. Build the model for downward load from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A 50 kg passenger stands on a scale in an elevator. Up is positive; use g=10 m/s². a (m/s²) -> N (N): (1.16, 558); (2.16, 608); (3.16, 658). Build the model for passenger on a scale from physics and predict both slope and intercept. **(3 pt)**